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Machine Learning

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Chapter one

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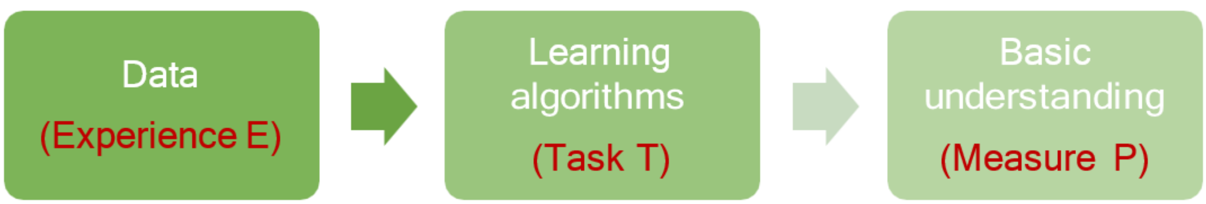
Machine learning is a core research field of AI, and it is also a necessary knowledge for deep learning. Therefore, this chapter mainly introduces the main concepts of machine learning, the classification of machine learning, the overall process of machine learning, and the common algorithms of machine learning.

1. Definition of Machine Learning

Machine learning (including deep learning) is a study of learning algorithms. A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E.

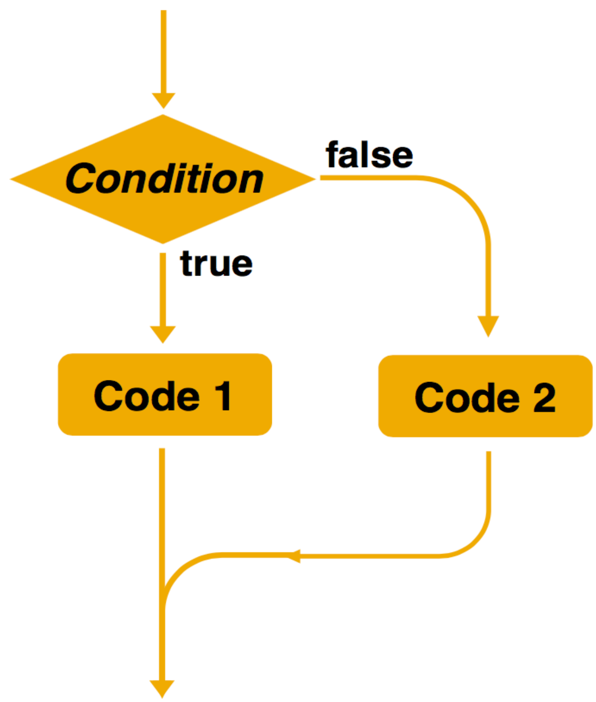


What this essentially means is that a machine can be considered to learn if it is able to gather experience by doing a certain task and improve its performance in doing the similar tasks in the future. When we talk about past experience, it means past data related to the task. This data is an input to the machine from some source. In the context of the learning to play checkers, E represents the experience of playing the game, T represents the task of playing checkers and P is the performance measure indicated by the percentage of games won by the player. The same mapping can be applied for any other machine learning problem, for example, image classification problem. In context of image classification, E represents the past data with images having labels or assigned classes (for example whether the image is of a class cat or a class dog or a class elephant etc.), T is the task of assigning class to new, unlabelled images and P is the performance measure indicated by the percentage of images correctly classified

2. Differences Between Machine Learning Algorithms and Traditional Rule-Based Algorithms

Rule-based algorithms

- Explicit programming is used to solve problems.
- Rules can be manually specified



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- Samples are used for training.
- The decision-making rules are complex or difficult to describe.
- Rules are automatically learned by machines

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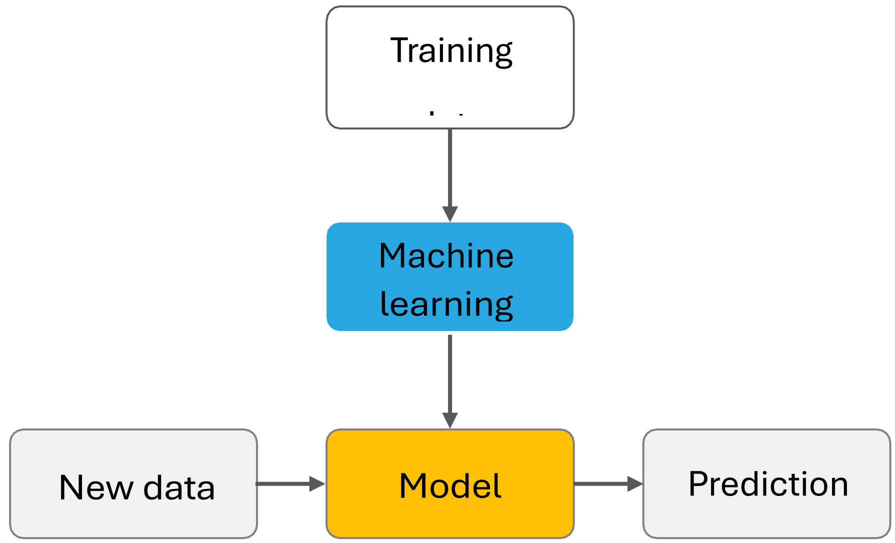


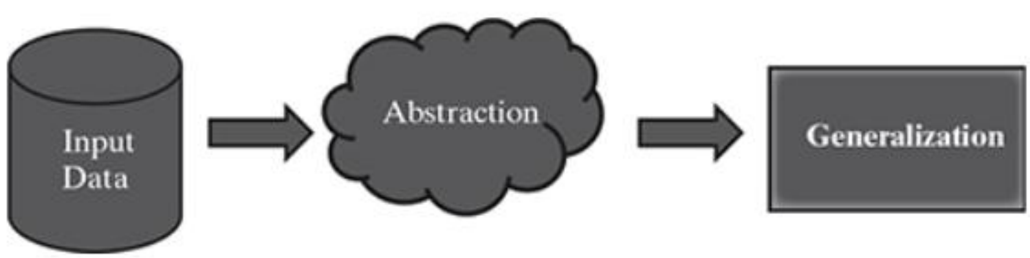
3. **How do machines learn**

The basic machine learning process can be divided into three parts.

1. Data Input: Past data or information is utilized as a basis for future decision-making

2. Abstraction: The input data is represented in a broader way through the underlying algorithm

3. Generalization: Generalization is the most important goal of machine learning. After learning patterns from training data, the model must apply this knowledge to new, unseen data.



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Let's consider the situation of typical process of learning from classroom and books and preparing for the examination. It is a tendency of many students to try and memorize (we often call it 'learn by heart') as many things as possible. This may work well when the scope of learning is not so vast. Also, the kinds of questions which are asked in the examination are pretty much simple and straightforward. The questions can be answered by simply writing the same things which have been memorized.

So, what we see in the case of human learning is that just by great memorizing and perfect recall, i.e. just based on knowledge input, students can do well in the examinations only till a certain stage. Beyond that, a better learning strategy needs to be adopted:

1. to be able to deal with the vastness of the subject matter and the related

issues in memorizing it

2. to be able to answer questions where a direct answer has not been learnt

A good option is to figure out the key points or ideas amongst a vast pool of knowledge. This helps in creating an outline of topics and a conceptual mapping of those outlined topics with the entire knowledge pool.

For example, a broad pool of knowledge may consist of all living animals and their characteristics such as whether they live in land or water, whether they lay eggs, whether they have scales or fur or none, etc. It is a difficult task for any student to memorize the characteristics of all living animals – no matter how much photographic memory he/she may possess. It is better to draw a notion about the basic groups that all living animals belong to and the characteristics which define each of the basic groups. The basic groups of animals are invertebrates and vertebrates. Vertebrates are further grouped as mammals, reptiles, amphibians, fishes, and birds. Here, we have mapped animal groups and their salient characteristics.

1. Invertebrate: Do not have backbones and skeletons

2. Vertebrate

1. Fishes: Always live in water and lay eggs

2. Amphibians: Semi-aquatic i.e. may live in water or land; smooth skin; lay eggs

3. Reptiles: Semi-aquatic like amphibians; scaly skin; lay eggs; cold-blooded

4. Birds: Can fly; lay eggs; warm-blooded

5. Mammals: Have hair or fur; have milk to feed their young; warm-blooded

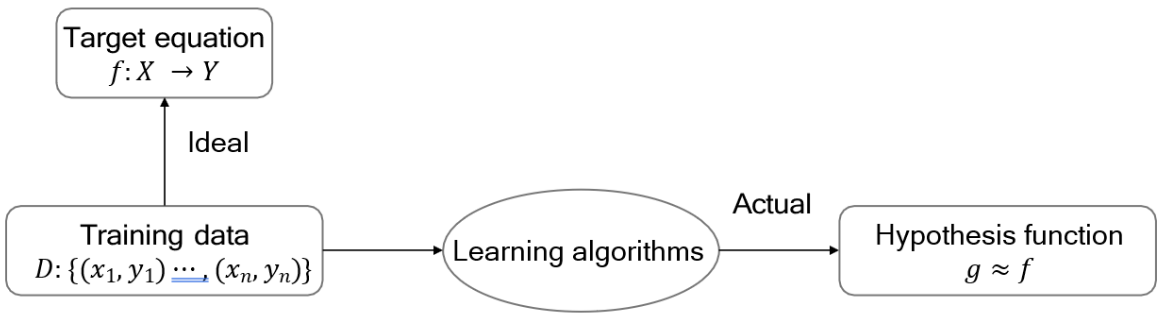
During the machine learning process, knowledge is fed in the form of input data. However, the data cannot be used in the original shape and form. As we saw in the example above, abstraction helps in deriving a conceptual map based on the input data. This map, or a **model** as it is known in the machine learning paradigm, is summarized knowledge representation of the raw data.

The **model** may be in any one of the following forms

- Computational blocks like if/else rules
- Mathematical equations
- Specific data structures like trees or graphs
- Logical groupings of similar observation

The first part of machine learning process is **abstraction** i.e. abstract the knowledge which comes as input data in the form of a model. However, this abstraction process, or more popularly training the model, is just one part of machine learning. The other key part is to tune up the **abstracted** knowledge to a form which can be used to take future decisions. This is achieved as a part of generalization. This part is quite difficult to achieve. This is because the model is trained based on a finite set of data, which may possess a limited set of characteristics.

4. **Rational Understanding of Machine Learning Algorithms**



- Target function ***f*** is unknown. Learning algorithms cannot obtain a perfect function ***f***.
- Assume that hypothesis function ***g* approximates** function ***f***, but may be different from function ***f***.

5. **Dataset**

Dataset: a collection of data used in machine learning tasks. Each data record is called a sample. Events or attributes that reflect the performance or nature of a sample in a particular aspect are called features.

Training set: a dataset used in the training process, where each sample is referred to as a training sample. The process of creating a model from data is called learning (training).

Test set: Testing refers to the process of using the model obtained after learning for prediction. The dataset used is called a test set, and each sample is called a test sample.

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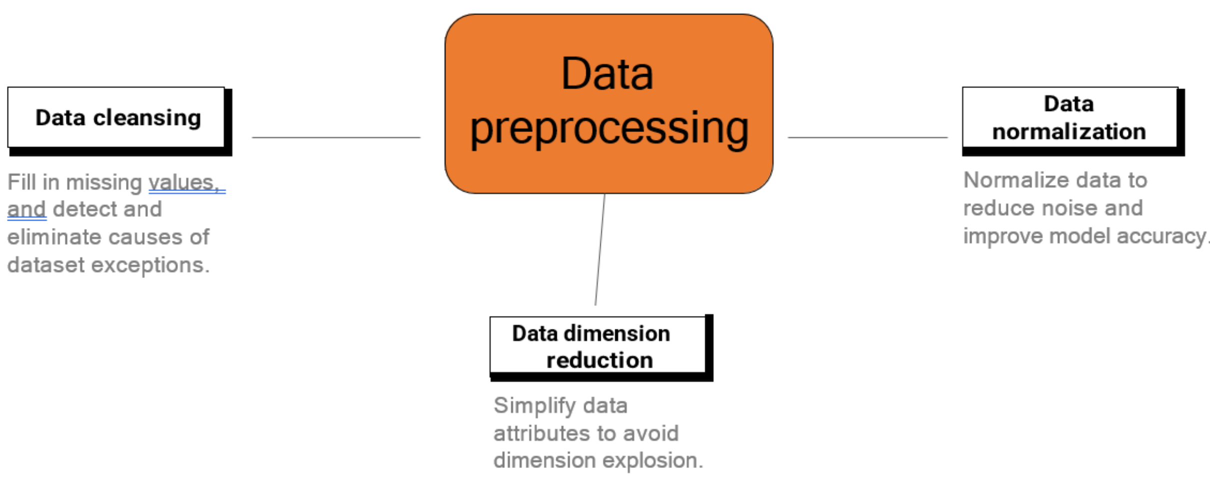
		Feature 1	Feature 2	Feature 3	Label
	No.	Area	School Districts	Direction	House Price
Training set	1	100	8	South	1000
	2	120	9	Southwest	1300
	3	60	6	North	700
	4	80	9	Southeast	1100
Test set	5	95	3	South	850

6. Data Processing

Data is crucial to models. It is the ceiling of model capabilities. Without good data, there is no good model.



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Data Cleansing

Most machine learning models process features, which are usually numeric representations of input variables that can be used in the model.In most cases, the collected data can be used by algorithms only after being preprocessed. The preprocessing operations include the following:

- Data filtering
- Processing of lost data
- Processing of possible exceptions, errors, or abnormal values
- Data consolidation(Combination of data from multiple data source)

Generally, real data may have some quality problems.

- Incompleteness: contains missing values or the data that lacks attributes
- Noise: contains incorrect records or exceptions.
- Inconsistency: contains inconsistent records.

Data Conversion

After being preprocessed, the data needs to be converted into a representation form suitable for the machine learning model. Common data conversion forms include the following:

- With respect to classification, category data is encoded into a corresponding numerical representation.
- Value data is converted to category data to reduce the value of variables (for age segmentation).
- Other data

- In the text, the word is converted into a word vector through word embedding (generally using the word2vec model, BERT model, etc).

- Process image data (color space, grayscale, geometric change, Haar feature, and image enhancement)

- Feature engineering
  - Normalize features to ensure the same value ranges for input variables of the same model.
  - Feature expansion: Combine or convert existing variables to generate new features, such as the average.

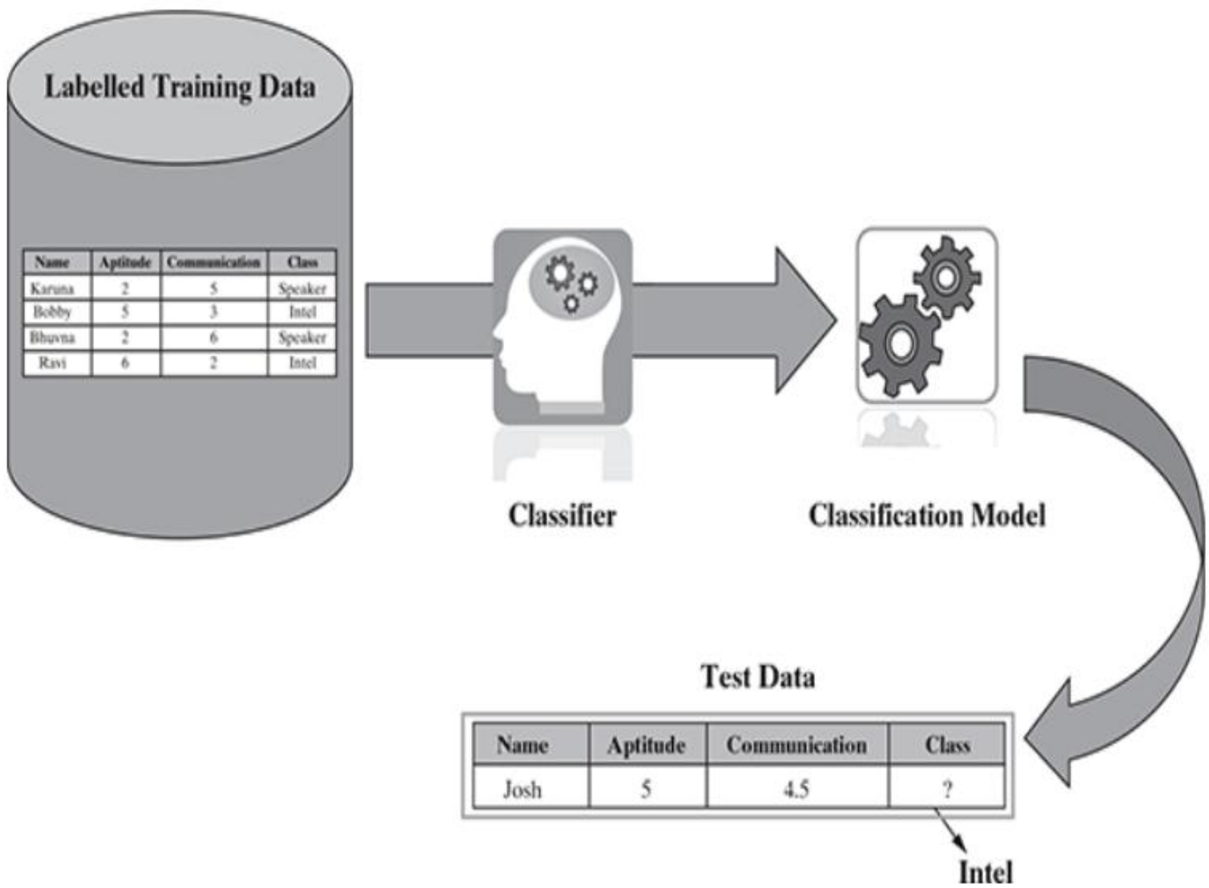
7. Machine Learning Classification

- **Supervised learning**: Obtain an optimal model with required performance through training and learning based on the samples of known categories. Then, use the model to map all inputs to outputs and check the output for the purpose of classifying unknown data. Supervised learning can be further classified into two types – Regression and Classification.

Classification: A computer program needs to specify which of the k categories some input belongs to. To accomplish this task, learning algorithms usually output a function $f: R^n \rightarrow (1,2,\ldots,k)$. such as analyzing positive/negative sentiment, male and female persons, benign and malignant tumors, secure and unsecure loans etc.

Some typical classification problems include:

- Image classification
- Prediction of disease
- Win–loss prediction of games
- Prediction of natural calamity like earthquake, flood, etc. Recognition of handwriting



Regression: For this type of task, a computer program predicts the output for the given input. Learning algorithms typically output a function $f: R^n \rightarrow R$. An example predicting real estate prices.

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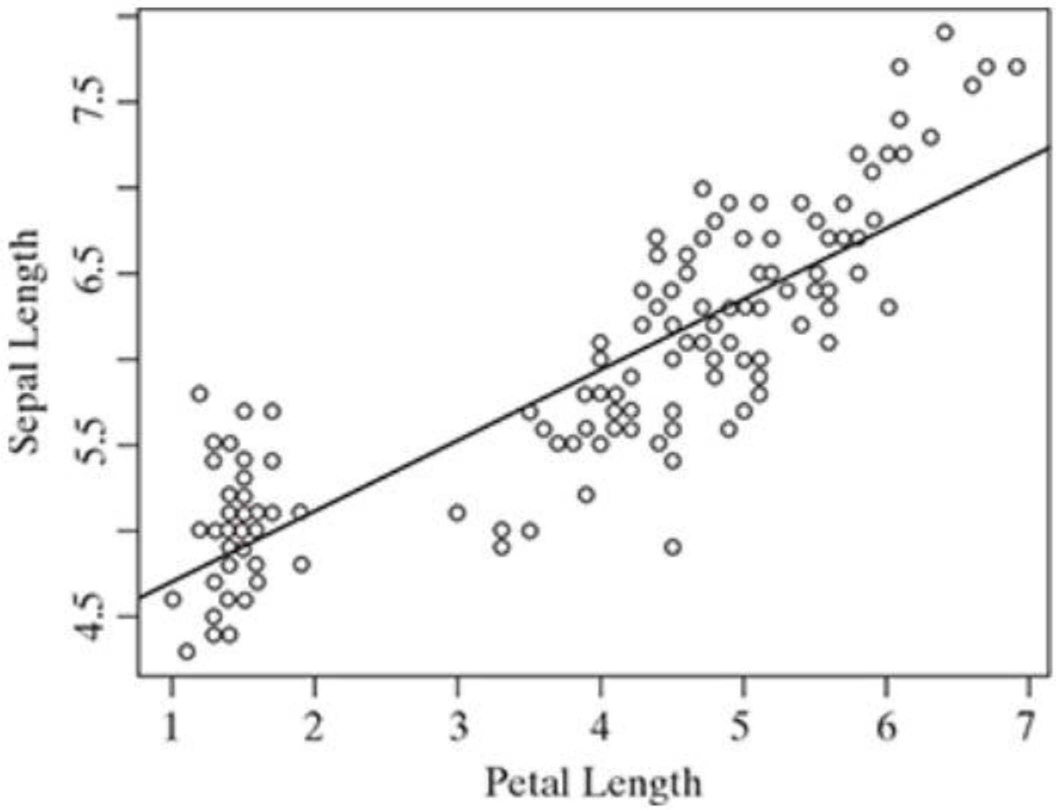
Figure shows a typical simple regression model, where regression line is fitted based on values of target variable with respect to different values of predictor variable. A typical linear regression model can be represented in the form

$$y = \alpha + \beta x$$

where 'x' is the predictor variable and 'y' is the target variable.



Typical applications of regression can be seen in

- Demand forecasting in retails
- Sales prediction for managers
- Price prediction in real estate
- Weather forecast

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- Skill demand forecast in job market

In supervised learning, learning data comes with description, labels, targets or desired outputs and the objective is to find a general rule that maps inputs to outputs. This kind of learning data is called **labeled data**. The learned rule is then used to label new data with unknown outputs.

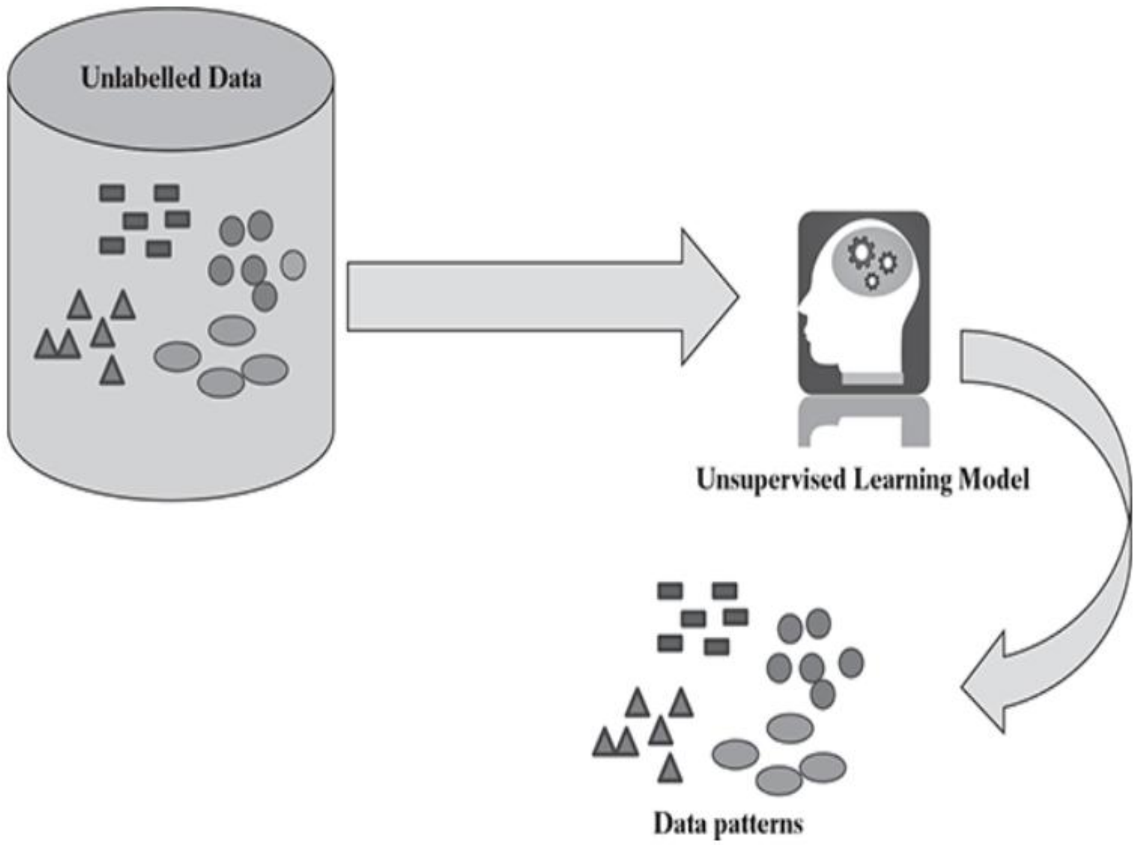
Supervised learning involves building a machine learning model that is based on **labeled samples**. For example, if we build a system to estimate the price of a plot of land or a house based on various features, such as size, location, and so on, we first need to create a database and label it. We need to teach the algorithm what features correspond to what prices. Based on this data, the algorithm will learn how to calculate the price of real estate using the values of the input features.

Supervised learning deals with learning a function from available training data. Here, a learning algorithm analyzes the training data and produces a derived function that can be used for mapping new examples. There are many **supervised learning algorithms** such as Logistic Regression, Neural networks, Support Vector Machines (SVMs), and Naive Bayes classifiers.

Common **examples** of supervised learning include classifying e-mails into spam and not-spam categories, labeling webpages based on their content, and voice recognition.

- **Unsupervised learning**: For **unlabeled** samples, the learning algorithms directly model the input datasets. Clustering is a common form of unsupervised learning. We only need to put highly similar samples together, calculate the similarity between new samples and existing ones, and classify them by similarity.

Clustering: A large amount of data from an unlabeled dataset is divided into multiple categories according to internal similarity of the data. Data in the same category is more similar than that in different categories. This feature can be used in scenarios such as image retrieval and user profile management.



- **Semi-supervised learning**: If some learning samples are labeled, but some other are not labeled, then it is semi-supervised learning. It makes use of a large amount of **unlabeled data for training** and a small amount of **labeled data for testing**. Semi-supervised learning is applied in cases where it is expensive to acquire a fully labeled dataset while more practical to label a small subset. For example, it often requires skilled experts to label certain remote sensing images, and lots of field experiments to locate oil at a particular location, while acquiring unlabeled data is relatively easy.

- **Reinforcement learning**: It is an area of machine learning concerned with how agents ought to take actions in an environment to maximize some notion of cumulative reward. The difference between reinforcement learning and supervised learning is the teacher signal. The reinforcement signal provided by the environment in reinforcement learning is used to evaluate the action (scalar signal) rather than telling the learning system how to perform correct actions.